INCH-POUND

MIL-C-48635 (AR)
AMENDMENT 4
02 November 1998
SUPERSEDING
AMENDMENT 3
12 November 1997

MILITARY SPECIFICATION

CARTRIDGE, 81MM, PRACTICE M879
LOADING, ASSEMBLING, AND PACKING

MIL-C-48635 was inactivated after 22 July 1996 for New Design

This amendment forms a part of MIL-C-48635 (AR), dated 02 July 1987, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

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2.1.2, Under DRAWINGS: Delete 9362798 and 9381730 in their entirety and substitute:

"12630600 - Packing and Marking for Container, PA 149, Ammunition Fiber for Cartridge, 81MM, M821E1 M889E1 and.M879.

12630601 - Packing and Marking for Box, Wirebound, for Cartridge, 81MM M879, M821E1 and M889E1."

Insert the following before "(Copies of specifications...)"

"TEST OPERATING PROCEDURES

TECOM TOP 3-2-825 Location of Impact or Airburst Positions."

AMSC N/A **FSC 1315**

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TABLE I PROVING GROUND TEST LINE: Delete “4.4.2.4” and substitute “4.4.3.1.1(A)”.

Add note (a):

“(a) A random sample of 96 cartridges shall be x-rayed and then sent to a proving ground for ballistic testing in accordance with 4.5.1.”

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4.4.2.3, Delete in its entirety and substitute:

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.3	TITLE Support, Propelling, Charge Assembly	SHEET 1 OF 1		DRAWING NUMBER 12977198
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY Inspection Method Reference
Critical	None defined			
Major				
101	Width of opening	(a)	3.1	Gage
102	Inside length	(a)	3.1	Gage
103	Inside radius to support wall	(a)	3.1	Gage
104	Radius to outside surface of rib (three places)	(a)	3.1	Gage
Minor				
201	Poor workmanship	0.65%	3.4	Visual

NOTES:
(a) Sample three (3) parts from each mold cavity at the start of each molding run.

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4.4.2.3.1: Delete in its entirety.

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4.4.2.4: Delete Critical 1 in its entirety and substitute:

"Major 113 – Depth of ignition cartridge from fin end below min. ... 100% ... 3.2 ... Gage

* 4.4.2.4: Delete Critical 3 in its entirety.

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4.4.2.4: Delete major defects 113, 114 and 115.

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4.4.2.3, Delete title and substitute:

"Packing and Marking for Container, PA 149, Ammunition for Fiber Cartridge, 81mm M821E1, M889E1 and M879 (while packing)"

Under Drawing Number: Delete "981730" and substitute "12630600".

Under Next Higher Assembly: Add "1263601".

Under Special A, after "Packing": Delete "Stop" and substitute "Clip".

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4.4.2.6, Delete title and substitute:

"Packing and Marking for Container, PA 149, Ammunition for Fiber Cartridge, 81mm M821E1, M889E1 and M879 (after packing)".

Under Drawing Number: Delete "981730" and substitute "12630600".

Under Next Higher Assembly: Delete "9362798" and substitute "12630601".

Major 102, under Examination or Test: Delete and substitute "Sealing strip (fiberglass) incomplete or badly wrinkled (before applying additional sealing strip (cloth)".

Major 103, under Examination or Test: Delete and substitute "Sealing strip (cloth) incomplete or badly wrinkled".

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4.4.2.7, Delete title and substitute:

"Packing and Marking for Box, Wirebound, for Cartridge, 81mm M879, M821E1 and M889E1".

Under Drawing Number: Delete "9362798" and substitute "12630361".

Minor 204, Delete in its entirety.

Add the following Minor defects:

"(Examinations while Packing before Sealing Box):

207 Box containing plastic sheet not appropriately marked ... 0.65% ... 3.1 ... Visual

208 Graphical firing scale not present in every third box ... 0.65% ... 3.1 ... Visual

209 Box containing graphical firing scale not appropriately marked ... 0.65% ... 3.1 ... Visual

210 Plastic sheet not present in every sixth box ... 0.65% ... 3.1 ... Visual"

4.4.2.8, Delete in its entirety and substitute:

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Cartridge, Practice, 81mm, M879 w/Fuze Practice, M751 (After Installation of Prop. Charge Sup. Ass'y)			9310340
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Support missing or damaged	0.40%	3.1	Visual
Minor 201	Poor workmanship	0.65%	3.4	Visual
NOTES:				

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4.4.3.1.1: Delete in its entirety and substitute:

"4.4.3.1.1 Sampling plan. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, ninety-six (96) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more short rounds occur.
c. One or more metal parts separations occur.
d. Horizontal range standard deviation at charge 0 exceeds 22 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
e. Three or more duds accumulatively occur in the charge 1 and charge 4 firings."

4.4.3.1.2: Delete in its entirety and substitute:

"4.4.3.1.2 Modified sampling plan. After three consecutive lots have complied with the criteria of 4.4.3.1.1, sixty (60) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. Two or more misfires occur.
b. One or more short rounds occur.
c. One or more metal parts separations occur.
d. Horizontal range standard deviation at charge 0 exceeds 22 meters or horizontal range standard deviation at charge 4 exceeds 83 meters.
e. Three or more duds accumulatively occur in the charge 1 and charge 4 firings."

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4.5.1: Delete in its entirety and substitute:

"4.5.1 Proving ground test. The test shall be performed at a government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges shall be fired from a standard 81mm M252 mortar at an elevation of forty-five (45) degrees and shall be temperature preconditioned to 70 degrees F plus or minus 5 degrees F for a minimum of twenty-four hours prior to firing. All rounds shall be fired onto a flat, nongrassy earthen surface that is not marshy or frozen. Twenty-four (24) cartridges or (twenty (20) cartridges for the modified test quantity) shall be fired at charge 0, twenty-four (24) cartridges or (twenty (20) cartridges for the modified test quantity) shall be fired at charge 1, and the remaining forty-eight (48) cartridges or (twenty (20) cartridges for the modified test quantity) shall be fired at charge 4. The charge 4 phase shall be fired before the charge 1 phase. Sixteen (16) data points are required to validate the charge 4 firing phase. If need be, up to four (4) additional cartridges may be diverted from the charge 1 phase test samples to the charge 4 test phase so that the minimum 16 data point requirement can be met. Individual velocities, pressures, times of flight, deflections, and ranges shall be recorded on the firing record for all charge phases (i.e., charge 0, charge 1 and charge 4). Duds for the charge 0 firings shall be recorded for information only. Range and deflection data for the charge 1 firings shall be recorded for information only. Observation shall be made for misfires and duds in all test phases and shall be recorded. The charge 0 phase shall be shot on a clean field. Each charge 0 impacted round shall be flagged and measured directly by optical methods. For both the charge 1 and charge 4 firing phases, three (3) downrange observers are required in the acquisition of range and deflection data. These observers shall be positioned with at least one beyond the impact zone and at least one short of the impact zone in such a manner that their downrange data can be tabulated in accordance with the procedures defined in TECOM 3-2-825, Appendix C. The fourth observer, as referred to in the TOP, is desired, but not required, for the charge 1 and charge 4 firing phases.

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After the first five (5) test rounds have been fired, the round impact data taken by the observers shall be used to validate their accuracy. If the reported data cannot be validated per the TOP, then those data shall be recorded, but not used in the downrange performance calculations. The observers shall then be required to analyze and correct the nature of their errors before firing can resume. In the event of a misfire, the ignition cartridge shall be removed and forwarded to ARDEC, ATTN: AMSTA-AR-FSA-MM for disassembly. Misfired ignition cartridges shall not be disassembled at the proving ground. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The muzzle velocities, range standard deviations and arithmetic averages for all rounds shall be calculated and recorded. Temperature of the day, weather conditions and the time of firing shall be recorded. Range probable error shall be calculated and recorded. Range probable error shall be calculated and recorded for all firing phases.

A smear camera shall be positioned to record each round in flight at about twenty (20) feet out of the muzzle. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all other rounds in that particular charge zone in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded."

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6.5, last two lines: Delete "Forty-eight samples thereafter" and substitute "Sixty (60) samples thereafter".

Add the following paragraphs:

"6.10 Marshy ground. Marshy ground is defined as an earthen surface that contains standing water or free flowing (fluid) mud or a combination of both.

6.11 Frozen ground. Frozen ground is defined as earthen ground within which water has frozen creating a hard surface."

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